

Cyber Security Basics

Chapter-1: Introduction

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2. The OSI Security Architecture
3. Security Attacks
4. Security Services
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What is Information ?

The processed form of data or **meaningful data** is called information. Basically, information is the message that is being conveyed.



What is Security ?

- Computer Security : Generic name for the collection of tools designed to protect data.
- Network and Internet Security : Measures to protect data during their transmission over a collection of interconnected networks.

Goal / Objective of Security

1. Confidentiality
2. Integrity
3. Availability
4. Authenticity
5. Accountability

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Confidentiality:

Data confidentiality: Assure confidential information not made available to unauthorized individuals.

Example : Individual files are locked and secured.

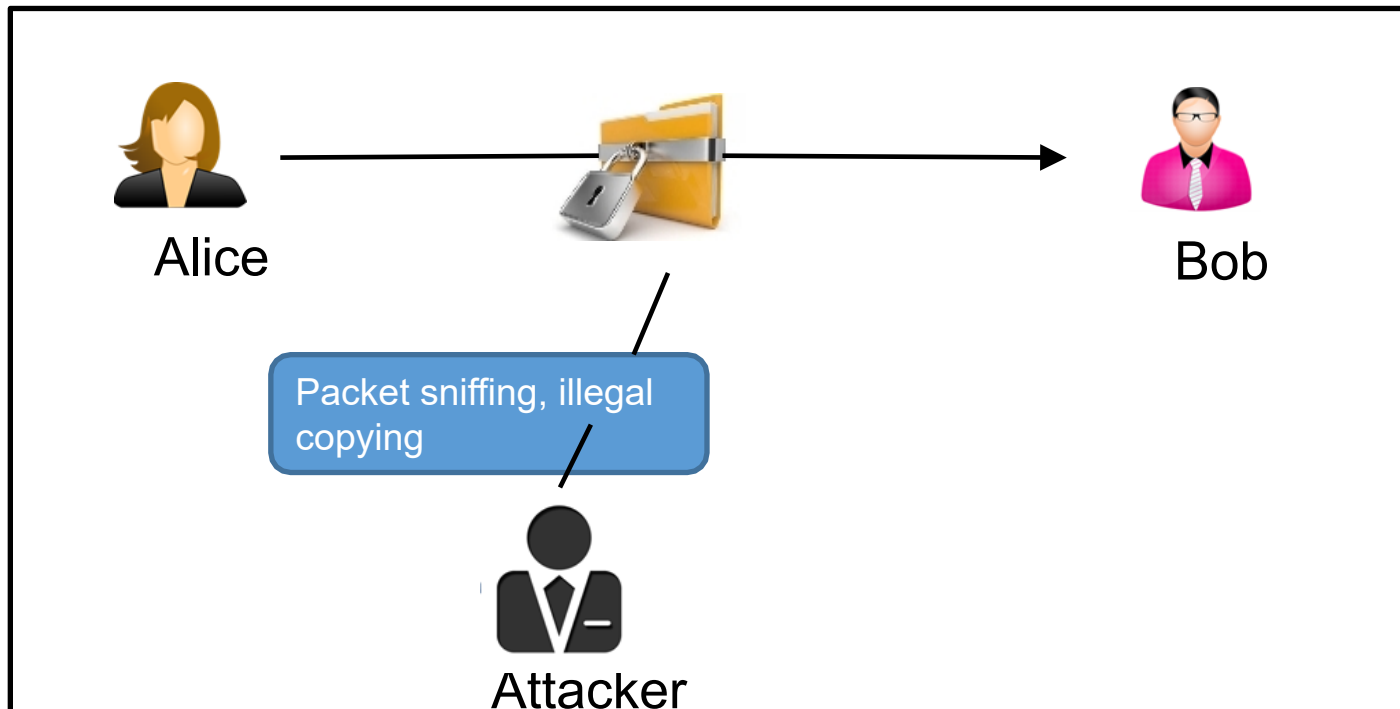


Fig 1.1 illustrating the concept of confidentiality

Confidentiality:

Privacy: Assure individuals can control what information related to them is collected, stored, distributed.

- Privacy is the right of an individual to protect personal or sensitive information.

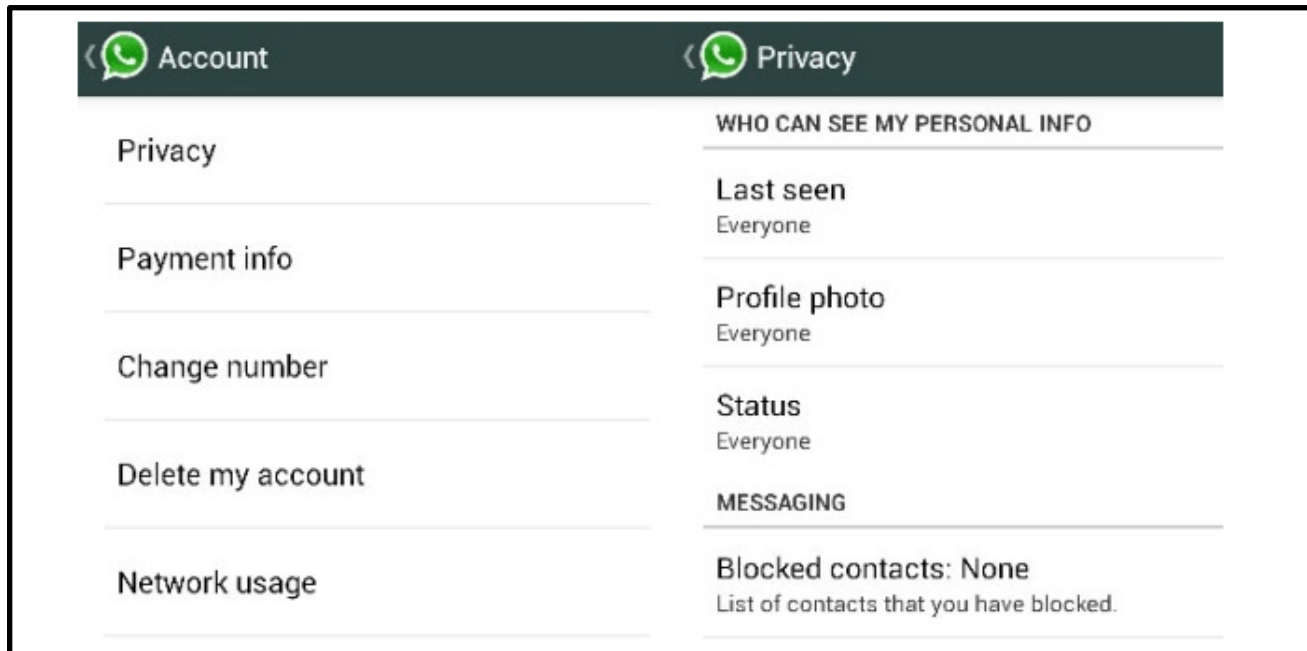


Fig 1.2 illustrating the concept of confidentiality

Integrity :

Integrity :

Data integrity: Assure information and programs are changed only in a authorized manner.

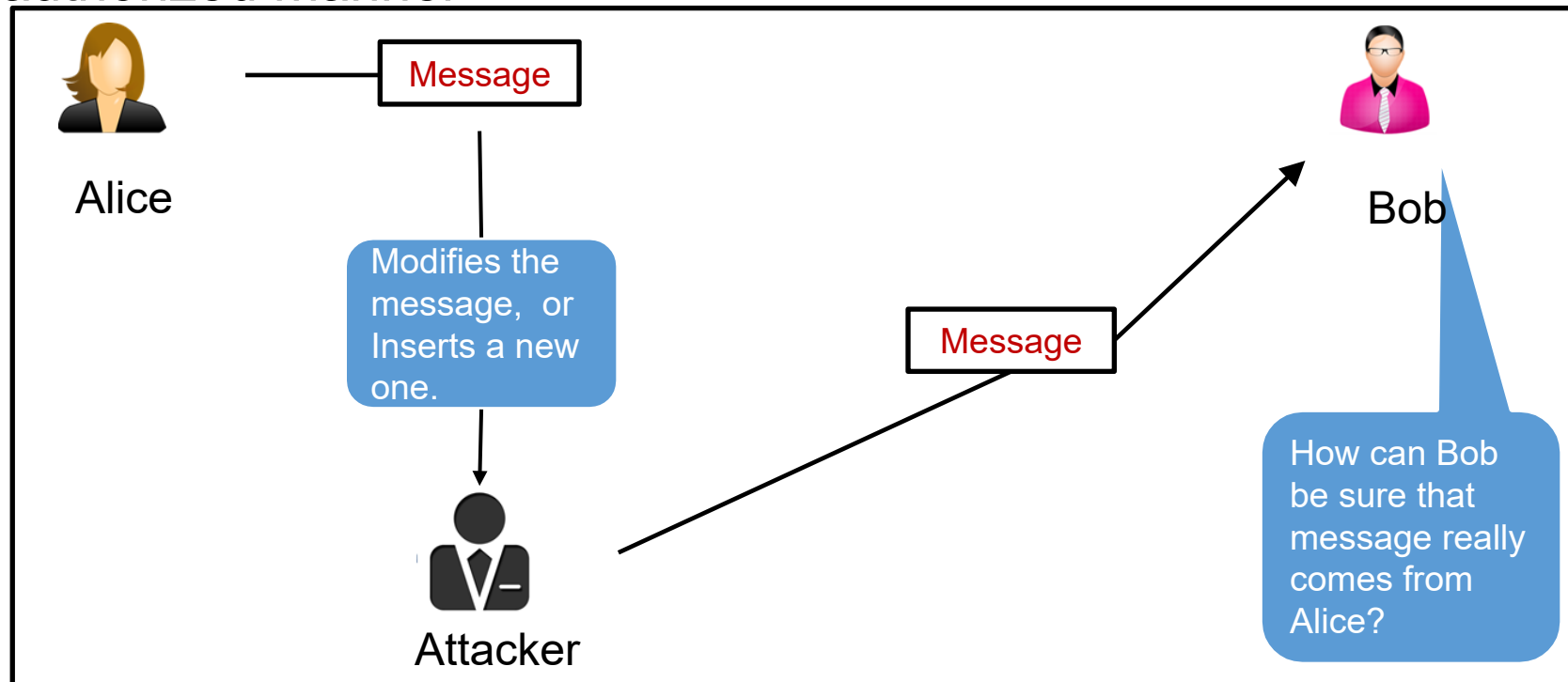


Fig 1.3 illustrating the concept of Integrity

Availability :

Assure that systems work promptly and service is not denied to authorized users.



Fig 1.4 illustrating the concept of Availability

Authenticity :

The property of being genuine and being able to be verified and trusted; confidence in the validity of a transmission, a message, or message originator.

This means verifying that each input arriving at the system came from a trusted source.

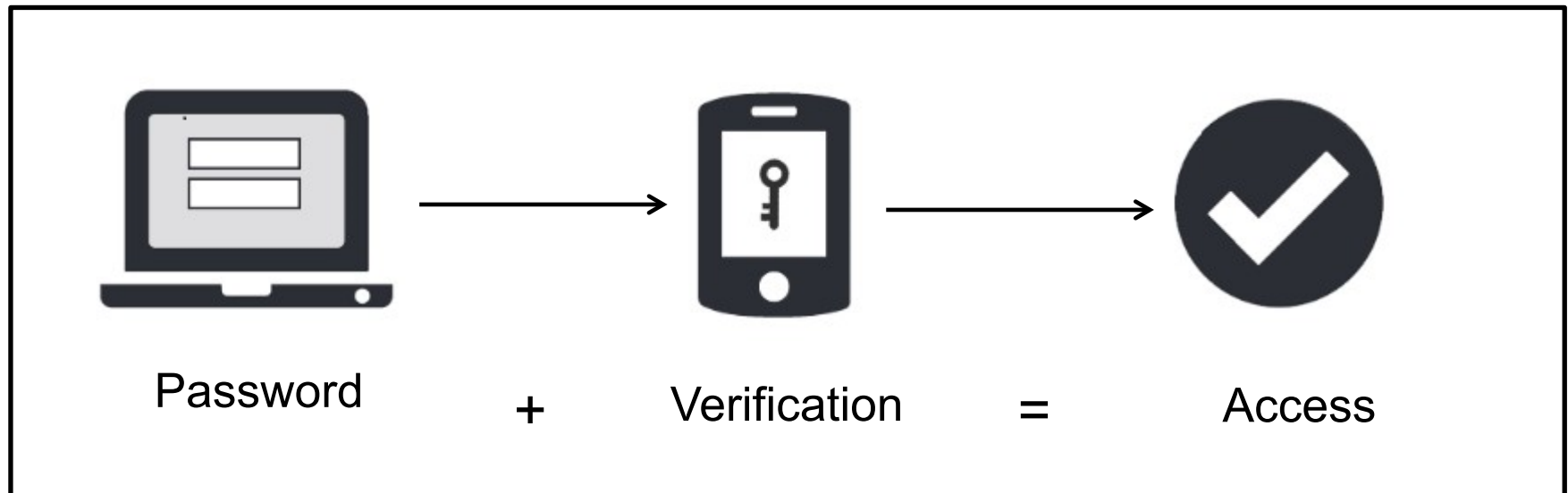


Fig 1.5 illustrating the concept of Authenticity

Authenticity :

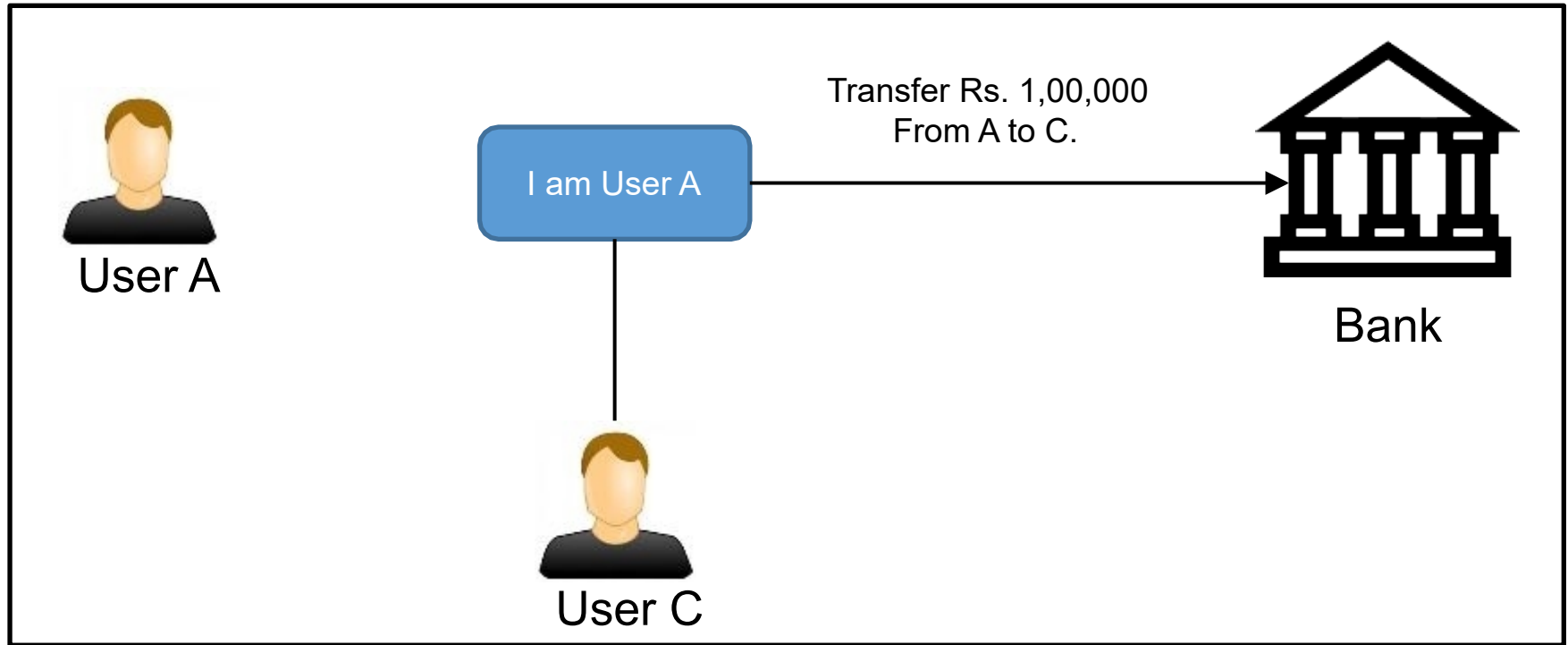


Fig 1.6 illustrating the concept of Authenticity

Accountability :

The security goal that generates the requirement for actions of an entity to be traced uniquely to that entity.

This supports nonrepudiation (assurance that someone cannot deny something).

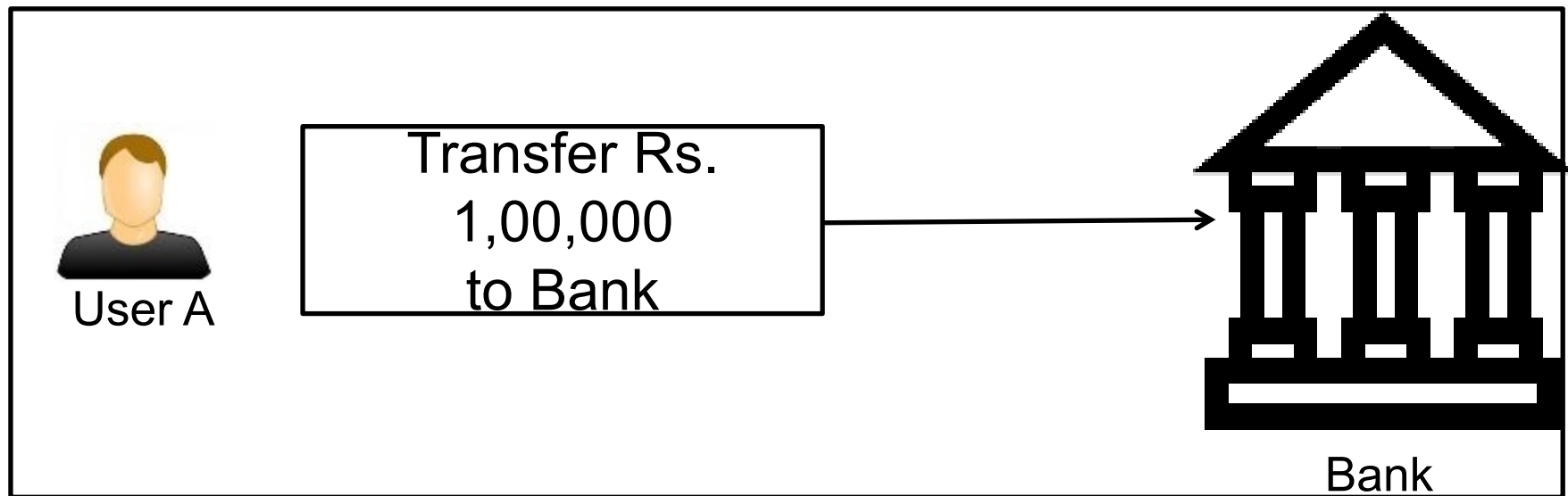


Fig 1.7 illustrating the concept of Accountability

Impact of Security Breach :

- Effectiveness of primary operations are reduced
Example: Hackers compromised exam question paper.
- Financial loss
Example: The cost of repairing a company database once it's been compromised.
- Damage to assets
Example: Hackers compromised Amazon Web Services account and demanded a ransom. When the company declined, the hacker started destroying their resources.
- Harm to individuals
Example: Hackers compromised patient database and change it.

Security Architecture:

Systematic approach to define requirements for security and approaches to satisfying those requirements

The OSI (Open Systems Interconnection) security architecture focuses on Security Attacks, Mechanisms, and Services.

Security Attack: Any action that compromises the security of information owned by an organization.

Security Mechanism: A process that is designed to detect, prevent, or recover from a security attack.

Security Service: A communication service that enhances the security of the data processing systems and the information transfers of an organization.

Security Attack:

A **passive attack** attempts to learn or make use of information from the system but does not affect system resources.

1. Release of message contents
2. Traffic analysis
 - Relatively hard to detect, but easier to prevent

An **active attack** attempts to alter system resources or affect their operation.

1. Masquerade
2. Replay
3. Modification of messages
4. Denial of service.
 - Relatively hard to prevent, but easier to detect

Security Attack – Passive Attack:

1. Release of Message content :

A telephone conversation, an electronic mail message, and a transferred file may contain sensitive or confidential information.

We would like to prevent an opponent from learning the contents of these transmissions.

Attack on Confidentiality.

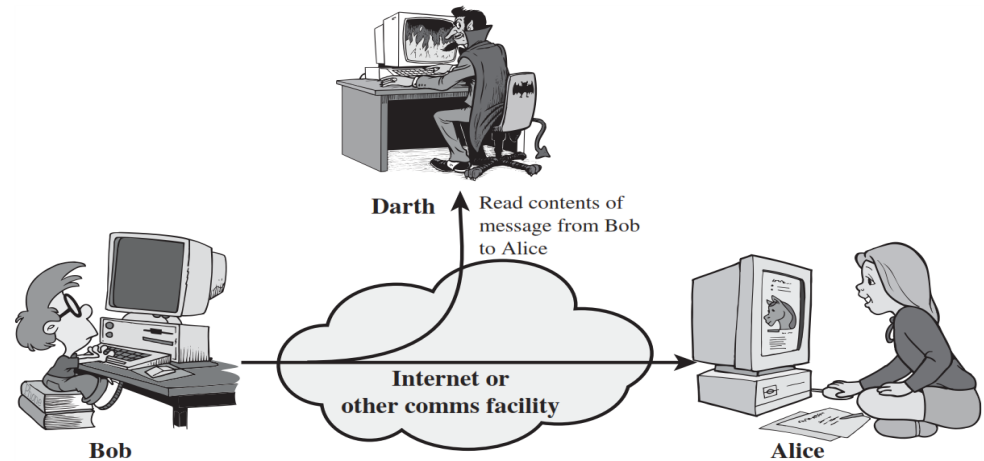


Fig 1.8 Release of Message Content

Security Attack – Passive Attack:

2. Traffic Analysis:

In such attacks, an attacker analyses the traffic and observes the frequency and length of exchanged messages. He uses all this information to predict the nature of communication.

Studying the flow of "traffic" to find patterns of behaviour.

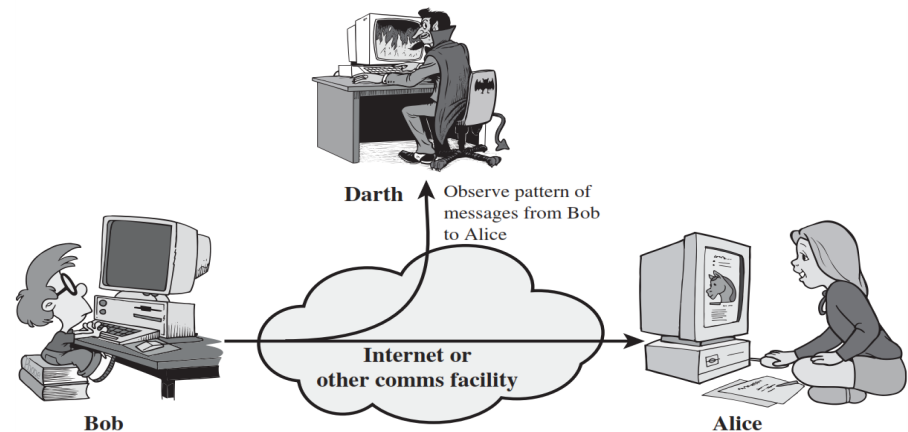


Fig 1.9 Traffic Analysis

Security Attack – Active Attack:

1. Masquerade Attack:

A masquerade takes place when one entity pretends to be a different entity.

A masquerade attack is an attack that uses a fake identity to gain unauthorized access to personal information.

Attack on Authentication.

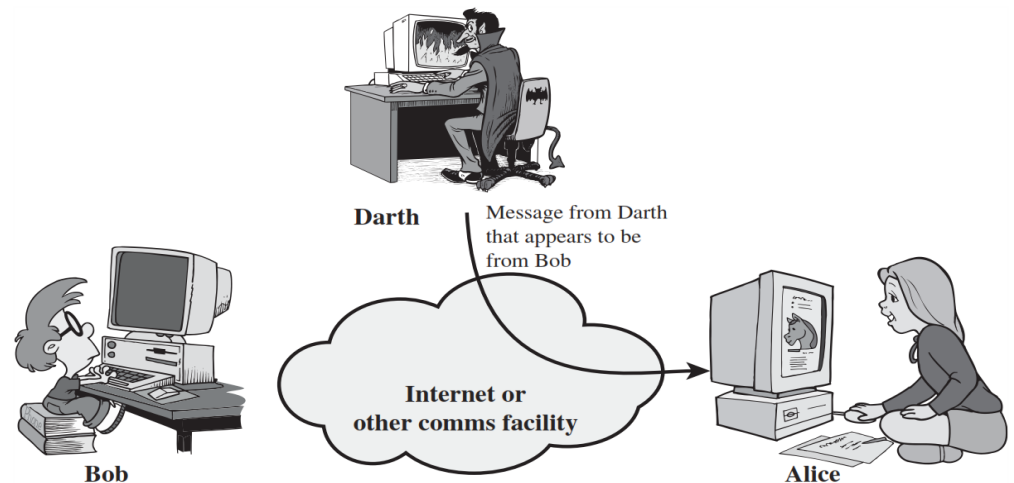


Fig 1.10 Masquerade Attack

Security Attack – Active Attack:

2. Replay Attack:

Replay attack involves the passive capture of a data unit and its subsequent retransmission to produce an unauthorized effect.

Replay attack is to replay the message sent to a network by an attacker, which was earlier sent by an authorized user.

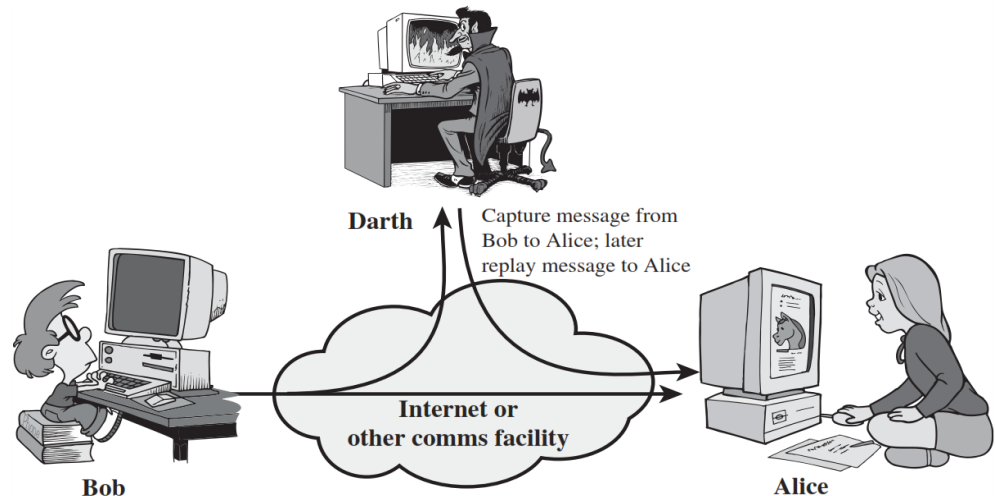


Fig 1.11 Replay Attack

Security Attack – Active Attack:

3. Modification of Message:

Modification of messages simply means that some portion of a legitimate message is altered, or that messages are delayed or reordered, to produce an unauthorized effect.

Attack on Integrity.

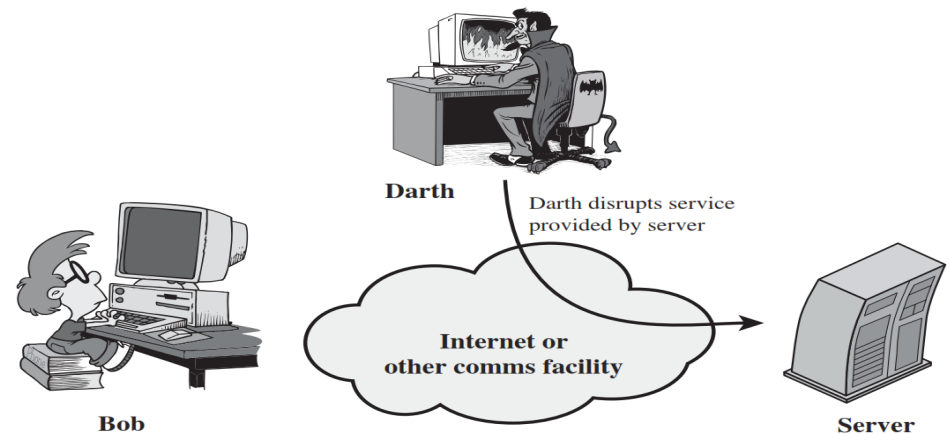


Fig 1.12 Modification of Message

Security Attack – Active Attack:

4. Denial of service Attack :

The denial of service attack prevents the normal use or management of communications facilities.

Sending large number of packets to block the server.

Attack on Availability.

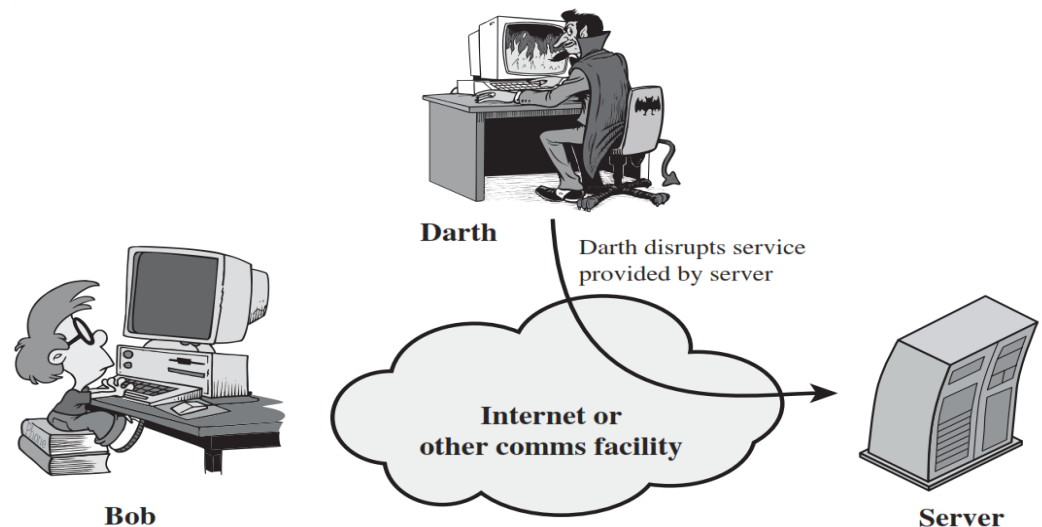
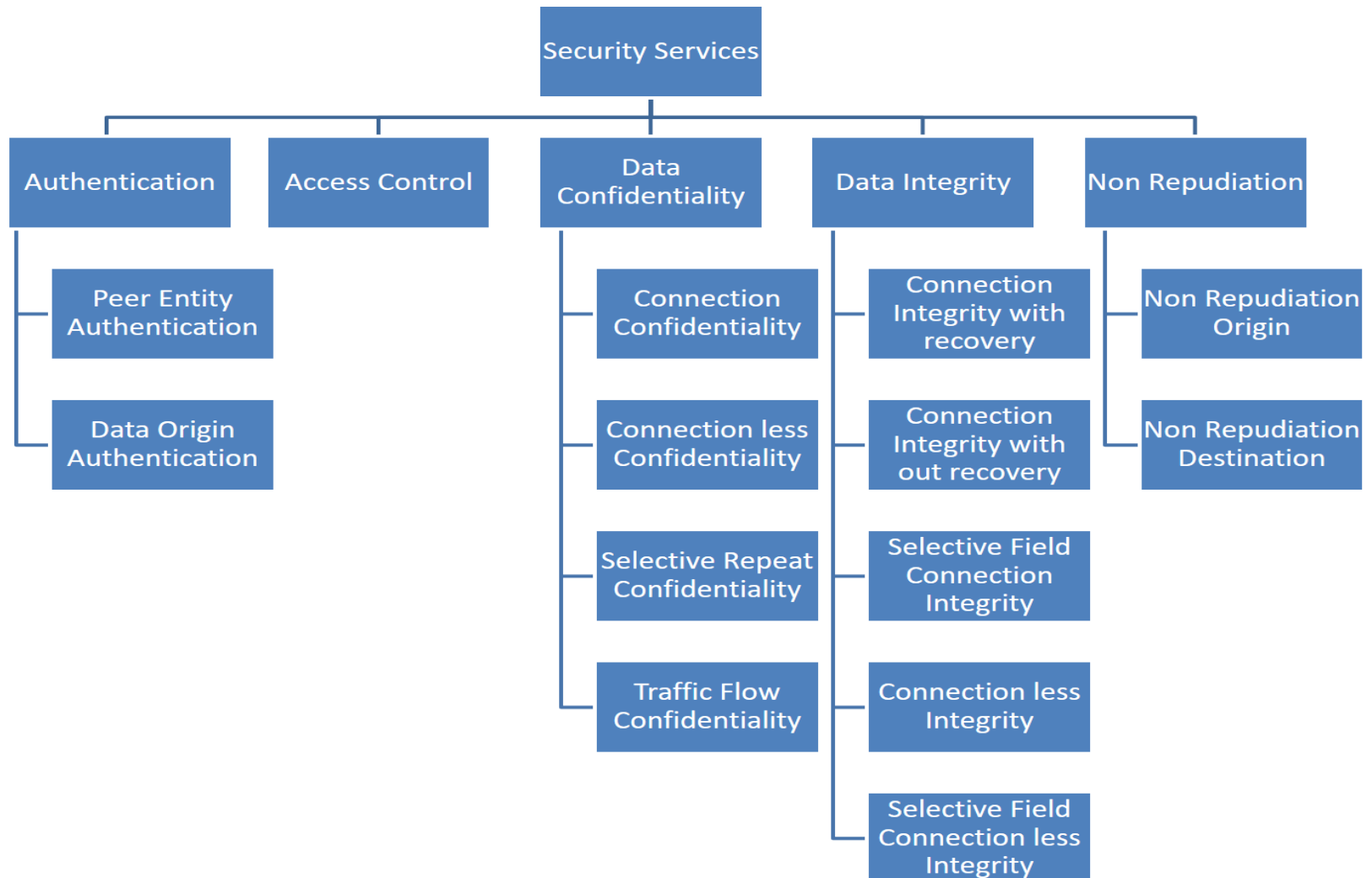


Fig 1.13 Denial of Service

Security Services:

- X.800 standard defines a security service as a service that is provided by a protocol layer of communicating open systems and that ensures security of the systems or of data transfers

Security Services:



Security Mechanism:

- Techniques designed to prevent, detect or recover from attacks
- No single mechanism can provide all services
- Common in most mechanisms: cryptographic techniques
- **Specific security mechanisms:** Integrated into the appropriate protocol layer in order to provide some of the OSI security services.
- **Pervasive security mechanisms:** Not integrated to any particular OSI security service or protocol layer

Security Mechanism :

- **Encipherment:** Hiding or covering data using mathematical algorithms.
- The sender can electronically sign the data and the receiver can electronically verify the signature.
- **Digital Signature:**
- **Access Control:** A variety of mechanisms that enforce access rights to resources.
- **Data Integrity:** A variety of mechanisms used to assure the integrity of a data unit or stream of data units.

Security Mechanism :

- **Authentication Exchange:** Two entities exchange some messages to prove their identity to each other.
- **Traffic Padding:** The insertion of bits into gaps in a data stream to frustrate traffic analysis attempts.
- **Routing Control:** Selecting and continuously changing routes between sender and receiver to prevent opponent(attacker) from eavesdropping.
- **Notarization:** The use of a trusted third party to assure and control the communication.

A model for Network Security :

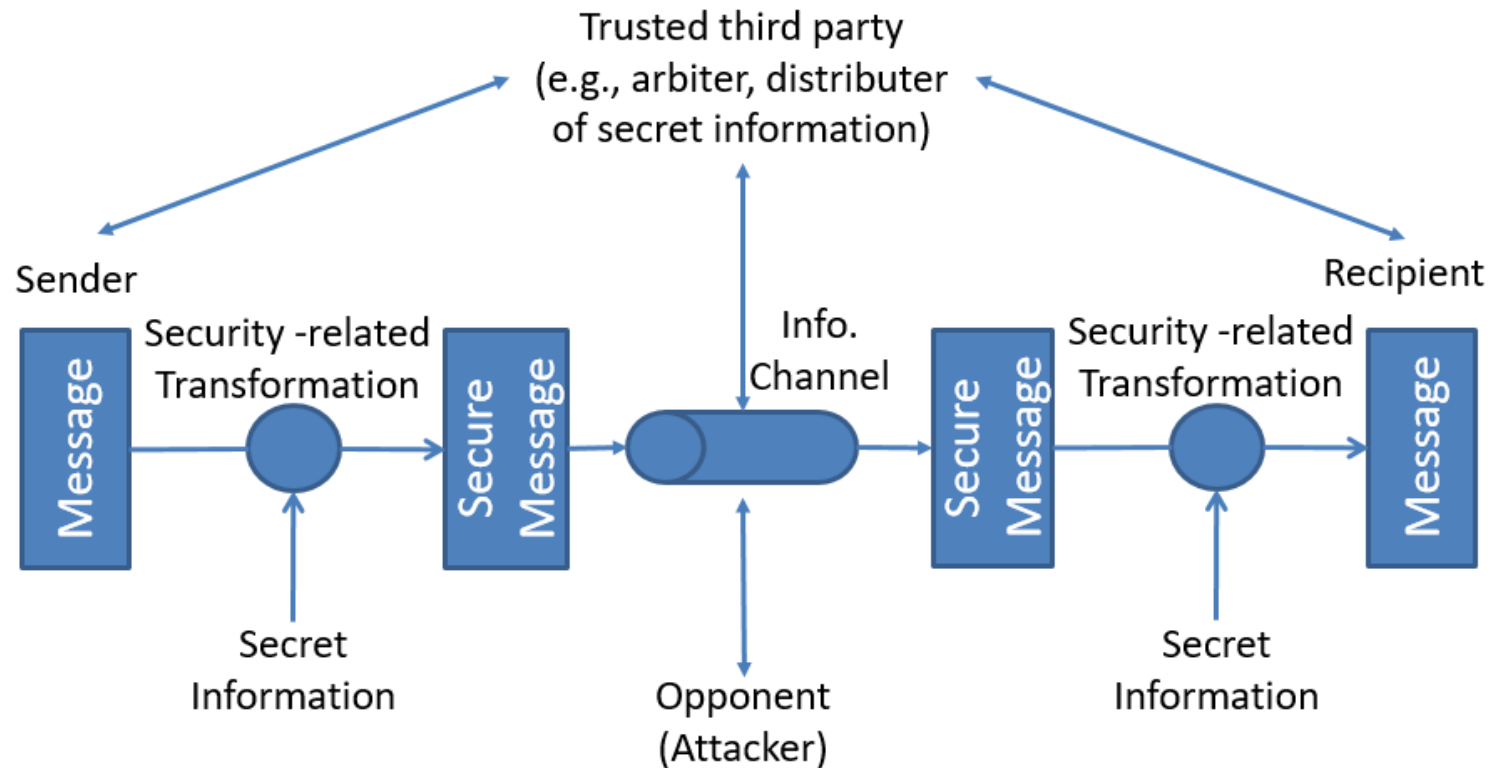
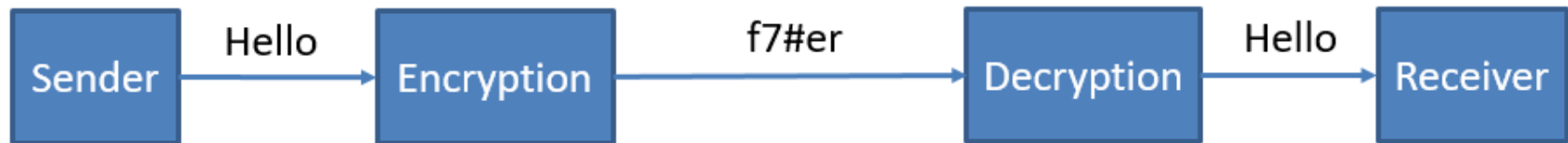


Fig 1.14 Model of Network Security

A model for Network Security :



A model for Network Security :

This image shows a general model of secure communication.

- On the left, there is the Sender, who has a Message.
- The message undergoes a Security-related Transformation (such as encryption), using some Secret Information (like a key). This produces a Secure Message.
- The secure message is then transmitted through an Information Channel.
- An Opponent (Attacker) may attempt to access or alter the message during transmission.
- At the receiving end, the Recipient gets the Secure Message.
- The secure message undergoes another Security-related Transformation (like decryption) using the Secret Information.
- Finally, the recipient recovers the original Message.

A model for Network Security :

- **Plaintext:** original message
- **Ciphertext:** encrypted or coded message
- **Encryption:** convert from plaintext to ciphertext (enciphering)
- **Decryption:** restore the plaintext from ciphertext (deciphering)
- **Key:** information used in cipher known only to sender/receiver
- **Cipher:** a particular algorithm (cryptographic system)
- **Cryptography:** study of algorithms used for encryption
- **Cryptanalysis:** study of techniques used for decryption without knowledge of plaintext
- **Cryptology:** areas of cryptography and cryptanalysis

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